

Restoration-Guided State-Space Adaptation for Degradation-Robust RGB Object Tracking

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Abstract—Visual object trackers can degrade under image-quality corruptions such as blur, low resolution, and sensor noise. This paper studies a conservative restoration-guided state-space adaptation of OTrack for RGB template-search tracking [1]. The final Paper Freeze V1 method inserts a Restoration-Guided State-Space Block (RG-SSB) into OTrack, keeps the backbone frozen, and trains only `rgssb.*` and `box_head.*` using degradation-aware LaSOT training with global clean-degraded feature consistency at weight $\lambda = 0.02$ [2]. Response consistency, target-region feature consistency, and target-versus-distractor margin loss are not retained in the final method. On selected benchmark subsets, the method improves average AUC on broader OTB by +0.033144 and expanded UAV123 by +0.017405, while corrected expanded NFS is slightly negative at -0.009638 . The cross-benchmark aggregate is near-neutral positive at +0.001478. These results support a cautious conclusion: restoration-guided feature adaptation shows robustness potential, but it does not establish state-of-the-art or universal degradation robustness.

Index Terms—Visual object tracking, degradation robustness, state-space models, Mamba, feature consistency, RGB tracking.

I. INTRODUCTION

RGB object tracking must maintain target localization under appearance change, motion, and image-quality degradation. Synthetic and real corruptions such as blur, low resolution, and Gaussian noise can weaken template-search matching, response-map localization, and bounding-box regression. Recent Mamba and state-space models have been explored in vision, while restoration-oriented variants such as MambaIR and MambaIRv2 adapt state-space modeling to image restoration [3], [4]. This paper evaluates a narrower question: whether restoration-oriented state-space feature adaptation can improve target-discriminative tracking features without changing the overall OTrack tracking architecture.

The goal is not to claim that Mamba is new to tracking, nor that degradation-robust tracking is unsolved. InvTrack is

direct local evidence that degradation-invariant RGB tracking can be formulated through clean/degraded training, feature consistency, and response-map fusion [5]. Instead, we study a specific missing intersection: restoration-oriented state-space feature adaptation inside RGB template-search tracking. The final method is intentionally conservative. It freezes the OTrack backbone, trains RG-SSB and the box head, and uses global clean-degraded feature consistency. This makes the empirical question testable while reducing the risk of conflating the proposed module with broad architecture changes.

The contributions of this Paper Freeze V1 manuscript are:

- a conservative OTrack+RG-SSB adaptation with frozen backbone and trainable RG-SSB plus box head;
- an evaluation under clean, motion-blur, low-resolution, and Gaussian-noise conditions on selected OTB, UAV123, and corrected NFS subsets;
- a corrected NFS provenance path that replaces invalid XYXY-as-XYWH NFS metrics with normalized aligned XYWH annotations;
- a same-GPU efficiency and parameter comparison on Tesla V100-PCIE-32GB hardware;
- a cautious claim boundary that reports positive OTB/UAV123 results, mixed corrected NFS transfer, and rejected negative ablations.

II. RELATED WORK

A. Template-Search Object Tracking

OTrack is used as the base tracker in this study [1]. The current paper does not propose a new tracking framework, temporal memory module, response fusion module, or template-update mechanism. Instead, it evaluates whether a restoration-guided state-space feature block can be inserted into the existing template-search pipeline while keeping the backbone frozen.

B. Degradation-Robust Tracking

InvTrack studies degradation-invariant single-object tracking with clean/degraded branches, feature consistency, and response-map fusion [5]. This is important prior evidence because it prevents overclaiming novelty for degradation-aware training alone. The present method differs by adapting a restoration-guided state-space block to tracking features rather than using the residual dense/low-pass residual design reported in the local InvTrack evidence card.

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C. Restoration-Oriented Mamba Models

MambaIR adapts state-space modeling to image restoration through residual state-space blocks with local enhancement and channel attention [3]. MambaIRv2 further studies attentive state-space restoration and semantic guided neighboring [4]. These methods address image restoration rather than template-search tracking. Our method borrows the restoration-guided feature-adaptation motivation but optimizes for tracking localization rather than image reconstruction.

III. PROPOSED METHOD

A. Overview

The final method uses OSTRack as the base tracker. Given a template image and a search image, OSTRack extracts transformer features and predicts a response map and bounding box. Paper Freeze V1 adds a Restoration-Guided State-Space Block (RG-SSB) after the backbone search features and before the tracking head. The module is designed to adjust tracking features under degradation rather than reconstruct RGB images.

The frozen method is:

- base tracker: OSTRack;
- added module: RG-SSB;
- frozen parameters: backbone;
- trainable parameters: `rgssb.*` and `box_head.*`;
- training data: degradation-aware LaSOT [2];
- feature consistency: global clean-degraded feature consistency;
- feature-consistency weight: $\lambda = 0.02$;
- disabled components: response consistency, target-region feature consistency, target-versus-distractor margin loss, degradation tokens, memory update, response fusion, and template-guided scan.

B. Training Objective

The training objective combines the degraded-branch tracking loss with global clean-degraded feature consistency:

$$\mathcal{L} = \mathcal{L}_{\text{track}}(X_d) + 0.02 \mathcal{L}_{\text{feat}}(F_c, F_d), \quad (1)$$

where X_d is the degraded search branch, and F_c and F_d are clean and degraded post-RG-SSB search features. The clean feature branch is detached in the consistency term. The final method intentionally keeps the response-consistency, target-region feature-consistency, and target-versus-distractor margin (TDM) objectives disabled.

IV. EXPERIMENTAL SETUP

A. Baseline and Final Configuration

The baseline is original OSTRack config `vitb_256_mae_ce_32x4_ep300`. The final method uses `vitb_256_mae_ce_32x4_ep300_rgssb_head_train_lasot_degraded_hpc_featcons_lam002`. All reported changes are computed as RG-SSB minus the OSTRack baseline for the same sequence, degradation, severity, and seed.

B. Benchmarks and Degradation Conditions

The evaluated subsets are:

- broader OTB: 13 sequences and 52 sequence-condition pairs [6];
- expanded UAV123: 16 sequences and 64 pairs [7];
- corrected expanded NFS: 32 sequences and 128 pairs [8].

Each sequence is evaluated under four conditions: clean, motion blur at medium severity, low resolution at medium severity, and Gaussian noise at medium severity. Degraded conditions use seed 42.

C. Metrics

The paper reports Success AUC, Precision@20, mean center error, model parameters, controlled FPS, per-frame latency, and peak allocated GPU memory. FLOPs/MACs are unavailable and are not reported as measured.

D. Corrected NFS Provenance

NFS required a correction before current results could be used. Raw NFS annotations were confirmed to contain `XYXY` coordinates, while the previous invalid path treated aligned rows as `XYWH`. The invalid NFS rows were archived, normalized aligned `XYWH` annotations were prepared, and NFS tracking was rerun. The current NFS metrics and failure analysis use only corrected normalized aligned `XYWH` annotations. Earlier NFS metrics and qualitative overlays are superseded.

V. RESULTS

A. Main Benchmark Results

Table I summarizes the benchmark-level results, and Fig. 1 visualizes the same average AUC changes. The strongest result is on OTB, where RG-SSB improves average AUC by $+0.033144$ across 52 sequence-condition pairs. UAV123 is also positive, with $+0.017405$ average AUC over 64 pairs. Corrected NFS is the main weakness: the method decreases average AUC by -0.009638 over 128 pairs. The combined cross-benchmark aggregate is $+0.001478$, which is best described as near-neutral positive rather than consistent superiority.

B. Condition-Level Results

Table II and Fig. 2 summarize condition-level behavior. The method is most consistently positive under motion blur. Low resolution remains the clearest condition-level weakness, with -0.008275 average AUC and increased center error.

C. Ablation Studies

Table III reports the final method-selection ablations, and Fig. 3 visualizes the decisions. Global feature consistency with $\lambda = 0.02$ is retained. Response consistency, target-region feature consistency, and TDM are rejected. TDM is important negative evidence: it improves several local sequence-condition pairs, but its Gaussian-noise regression dominates the aggregate gate result. It is therefore not part of the frozen method.

The final TDM gate decision is `REJECT_AND_ROLL_BACK_TO_PAPER_FREEZE_V1`.

TABLE I
PAPER FREEZE V1 BENCHMARK SUMMARY. POSITIVE AUC AND PRECISION@20 CHANGES FAVOR RG-SSB; NEGATIVE CENTER-ERROR CHANGES INDICATE LOWER ERROR.

Benchmark	Sequences	Pairs	Baseline AUC	RG-SSB AUC	AUC Change	Precision Change	Center Error Change	Conclusion
OTB	13	52	0.646964	0.680108	+0.033144	+0.050250	-4.563202	Improves clearly on average
UAV123	16	64	0.660183	0.677587	+0.017405	+0.029983	-3.500185	Improves slightly / near neutral
NFS	32	128	0.681149	0.671511	-0.009638	-0.006937	+3.705562	Mixed / negative average
Cross-benchmark aggregate	53	212	0.689269	0.690748	+0.001478	+0.007843	+1.161416	Near-neutral positive overall

TABLE II
CONDITION-LEVEL ROBUSTNESS SUMMARY ACROSS SELECTED OTB, EXPANDED UAV123, AND CORRECTED EXPANDED NFS SUBSETS.

Condition	Pairs	Baseline AUC	RG-SSB AUC	AUC Change	Precision Change	Center Error Change	Conclusion
Clean	61	0.685280	0.690648	+0.005368	+0.015220	+2.558046	Improves on average
Motion blur	61	0.643687	0.663299	+0.019612	+0.029967	-7.502857	Improves on average
Low resolution	61	0.672450	0.664175	-0.008275	-0.003989	+6.769353	Mixed / negative
Gaussian noise	61	0.672042	0.681627	+0.009585	+0.018540	-0.342091	Improves on average

TABLE III
ABLATION SUMMARY FOR PAPER FREEZE V1. LOCAL ABLATIONS GUIDE METHOD SELECTION AND ARE NOT FULL BENCHMARK CLAIMS.

Ablation	Scope	Primary Comparison	Average AUC Change	Decision	Evidence
Global feature consistency $\lambda = 0.02$	3 OTB sequences $\times$ 4 conditions	vs. OSTRack baseline	+0.023480	Kept	Lambda sweep CSV
Response consistency	3 OTB sequences $\times$ 4 conditions	vs. feature-only $\lambda = 0.02$	-0.019360	Rejected	Response-consistency CSV
Target-region feature consistency	3 OTB sequences $\times$ 4 conditions	vs. global feature consistency	-0.023555	Rejected	Target-vs-global CSV
Target-vs-distractor margin	3 OTB sequences $\times$ 4 conditions	vs. global feature consistency	-0.014549	Rejected by gate	TDM gate CSV

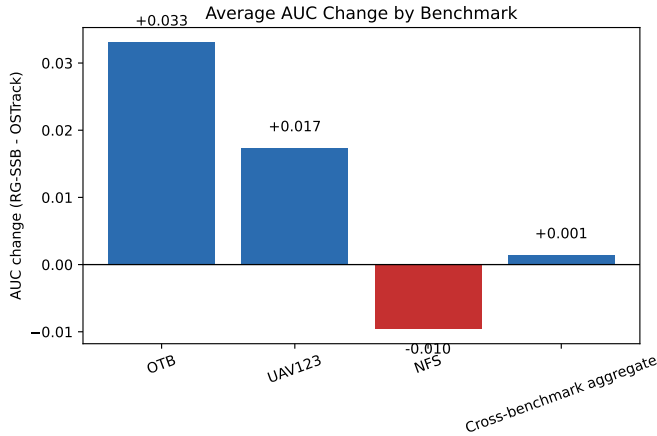


Fig. 1. Benchmark-level AUC change for Paper Freeze V1. Positive values favor RG-SSB. The evaluated scope is selected OTB, expanded UAV123, corrected expanded NFS, and the cross-benchmark aggregate.

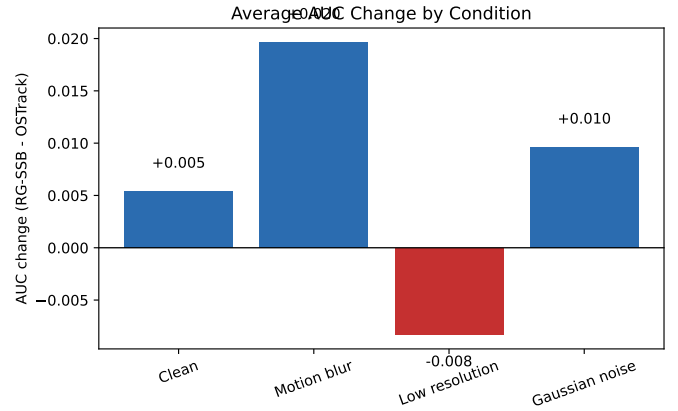


Fig. 2. Condition-level AUC change across clean, motion blur, low resolution, and Gaussian noise. The low-resolution condition remains mixed/negative.

D. Efficiency Analysis

Table IV and Fig. 4 report the same-GPU controlled efficiency comparison. The measurement uses a Tesla V100-PCIE-32GB, the same Car1 clean sequence, 50 warm-up frames, 3 repetitions, synchronized CUDA timing, and separate initialization/per-frame tracking measurements. Historical timing files are not mixed with this controlled comparison.

The final method adds 2,079,936 parameters over the baseline. FLOPs/MACs are unavailable and are not reported as measured.

E. Corrected NFS Failure Analysis

Figure 5 summarizes terminal outcomes for 10 corrected NFS failure cases selected from the largest corrected drops. Persistent RG-SSB-specific target loss appears in 5 cases, sudden center jump in 3 cases, shared tracker failure in 1 case, and ambiguous behavior in 1 case. This is a qualitative failure-focused analysis, not a benchmark-wide frequency estimate.

VI. DISCUSSION AND LIMITATIONS

The evidence supports a narrow, defensible claim: restoration-guided state-space adaptation can improve selected robustness outcomes in RGB template-search tracking, especially on the tested OTB subset. UAV123 is positive but mixed, and corrected NFS is slightly negative. The result is therefore promising but not uniformly improved.

TABLE IV
CONTROLLED SAME-GPU EFFICIENCY COMPARISON ON TESLA V100-PCIE-32GB. FLOPS/MACs WERE UNAVAILABLE AND ARE NOT REPORTED AS MEASURED.

Model	Params	Trainable Params	Frozen Params	Init. ms	Latency ms	FPS	Peak Alloc. MB	Peak Reserved MB	FLOPs/MACs
OTrack baseline	92,518,533	92,518,533	0	2.949	10.724	93.247	380.233	434.000	Unavailable
RG-SSB final	94,598,469	8,554,053	86,044,416	3.165	11.525	86.766	388.917	434.000	Unavailable

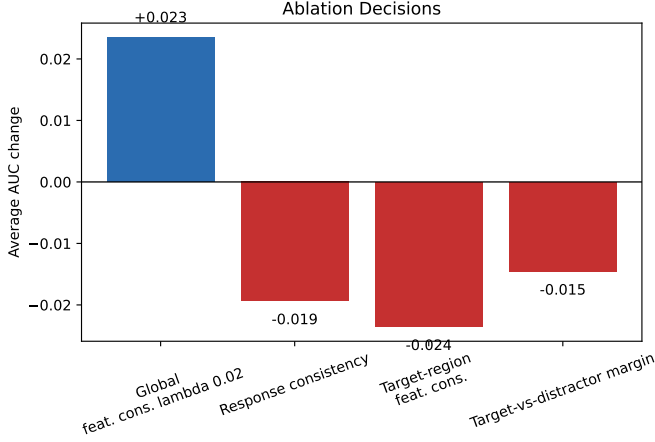


Fig. 3. Ablation decision summary. TDM is reported only as rejected negative evidence and is not part of Paper Freeze V1.

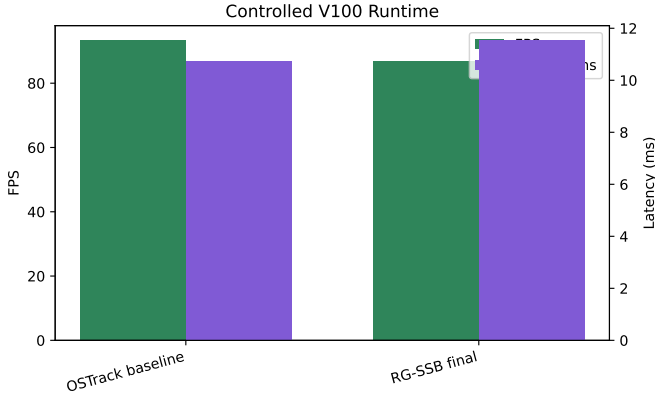


Fig. 4. Controlled V100 runtime comparison. The final method has a modest FPS reduction relative to the baseline and a small increase in peak allocated memory.

The low-resolution weakness and corrected NFS failures are important limitations. They suggest that global feature consistency may not fully protect target identity under fast motion, high-frame-rate sampling behavior, or severe target/background ambiguity. The rejected TDM ablation further shows that adding a response-margin objective is not automatically beneficial; poorly balanced response constraints can introduce large condition-specific regressions.

This paper version has several boundaries. The benchmark coverage uses selected subsets, not complete benchmark suites. Degradations are synthetic, medium severity, and use one seed. Training uses LaSOT only. The backbone remains frozen. FLOPs/MACs are unavailable. Corrected NFS required annotation normalization and rerunning NFS tracking; older NFS

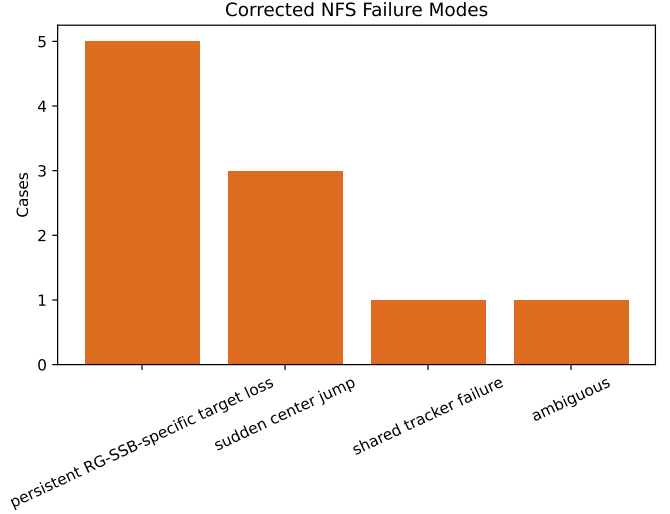


Fig. 5. Corrected NFS failure-mode distribution for 10 selected failure cases. Ground truth and diagnostics use corrected aligned XYWH annotations.

metrics and overlays are superseded. The evidence does not support state-of-the-art claims, universal degradation robustness, or uniform gains across all benchmarks.

VII. CONCLUSION

Paper Freeze V1 establishes a conservative fallback method: OTrack with RG-SSB and box-head training under global clean-degraded feature consistency. The method is positive on OTB and UAV123, mixed on corrected NFS, and near-neutral positive in the mandatory cross-benchmark aggregate. This is sufficient for a cautious proof-of-concept paper position, but not for claims of broad superiority. Future work should expand benchmark coverage, test multiple seeds and severities, and investigate failure-specific refinements only after the current evidence is presented with appropriate limits.

APPENDIX A REPRODUCIBILITY NOTES

The final method config is `vitb_256_mae_ce_32x4_ep300_rgssb_head_train_lasot_degraded_hpc_featcons_lam002`. The frozen method keeps the backbone frozen and trains only `rgssb.*` and `box_head.*`. NFS results must use the corrected normalized aligned XYWH annotation root. The invalid historical NFS rows are archived for provenance only and must not be used for current paper claims. FLOPs/MACs are unavailable and should not be reported as measured.

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